

Predicting Fourth Quarter Running Performance of Elite Male Gaelic Footballers from GPS Match Data

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Abstract

This study presents the first application of machine learning to predict fourth quarter (Q4) high-speed running (HSR) performance in elite Gaelic football from pre-Q4 Global Positioning System (GPS) movement data. Using 99 player-game observations from an anonymised inter-county team across 11 matches and 3 seasons, four rounds of models were evaluated: ARIMA, BiLSTM with attention, hand-crafted tabular models, and TSFresh automated feature extraction, across five training configurations that systematically vary the availability of personal player history. All learned models outperform naive baselines, confirming that pre-Q4 movement patterns carry genuine predictive signal. Ridge regression with TSFresh features and a historical Q4 average achieves the best regression performance (MAE 4.721 m·min⁻¹), while a BiLSTM with attention achieves the strongest classification result (F1 0.763). A refined four-feature Ridge model achieves near-equivalent performance to the best automated model while remaining fully explainable to a practitioner. Results suggest that player profile consistency matters more than volume of historical data, and that interpretable tabular models are the recommended practical configuration for pitch-side deployment.

Keywords: Gaelic football · GPS tracking · machine learning · high-speed running · fourth quarter prediction · time series · player performance

1 Introduction

Gaelic football is an Irish field invasion sport contested at elite level by inter-county teams across 34 counties [1]. The physical demands for players are substantial and constantly evolving due to advances in strength and conditioning and quicker transitional play [2], resulting in increased running loads across games. Players typically cover between 7,000 and 10,000 metres per game, with 1,300 to 2,200 metres at high speed [3]. Performance metrics such as total dis-

tance, high-speed running and sprint distance, captured by GPS data, have become primary indicators of physical output at inter-county level [2]. While this data can be used to effectively determine player workload, it is almost exclusively used reactively to understand what occurred during a game, rather than predicting what will transpire next.

A consistent finding in the literature is that the fourth quarter is the most decisive phase of a game. Gamble et al. recognised that teams maintaining their running output in Q4 have significantly higher winning probabilities, highlighting the correlation between Q4 running performance and game outcome [4]. Malone et al. also demonstrated that high-speed running distance declines dramatically in Q4 relative to Quarter 1 (Q1), with a steeper decline observed in losing teams [5]. Despite these findings, no published work has attempted to predict Q4 running performance before the final quarter commences. Existing GPS analysis tools provide only real-time measurement with no predictive capability, including StatSports Sonra, the most widely used tool at inter-county level [6]. Forecasting whether a player’s Q4 high-speed running output will improve or decline has the ability to identify underperforming players, support substitution decisions and manage workload at the most consequential moment of a game [7].

This study addresses that gap, shifting from descriptive analysis to focus on predictive modelling. Using GPS data from an anonymised inter-county team across 11 matches and 3 seasons [8, 9], we investigate three questions: whether pre-Q4 movement patterns can predict Q4 high-speed running performance, whether access to a player’s historical Q4 data improves those predictions, and which features and movement zones drive model decisions. Two prediction targets are defined: Q4 HSR per minute as a continuous regression target, and a binary classification label, Improve or Decline, derived from whether Q4 output increases or decreases compared to the player’s pre-Q4 output average. The choice of Q4 HSR metres per minute as the primary prediction target is further motivated by Mangan et al. establishing that players maintaining higher HSR in Q4 also retain greater technical accuracy [3], reinforcing HSR as a representative for overall playing quality [10, 11].

An experimental question beyond predictive performance is the role of player identity. Three core training configurations are evaluated: training on all other players only, representing a cold-start scenario where the target player has no personal data in the training set (the model must rely entirely on population-level patterns from other players), training on the target player’s own prior games representing full personalisation, and combining both configurations. To extend this, two further configurations explored cluster-based and upweighted training. This design tests whether personalised models outperform population-level approaches and how prediction varies with available player history. Two feature paradigms are compared: five interpretable hand-crafted summary statistics derived from the pre-Q4 rolling window sequence [12], and automated features extracted from TSFresh [13]. A BiLSTM with attention is also trained directly

on the raw 50-step multivariate sequences [14], learning temporal patterns across all 6 movement zones without manual feature engineering.

We hypothesise that pre-Q4 movement patterns carry genuine predictive signal for Q4 running performance, that access to a player’s historical Q4 data will substantially improve accuracy for both regression and classification, and that interpretable tabular models will outperform neural network models given the limitations of a 99-observation dataset.

1.1 Contributions

- First application of machine learning to predict Q4 HSR performance in Gaelic football, shifting from descriptive post-match analysis to prospective within-game prediction.
- Systematic comparison of five training configurations providing a structured evaluation of how data availability and player identity affect prediction quality across cold-start and personalised scenarios.
- Automated feature extraction adds minimal value over four interpretable features that can be explained to a coach in one sentence, with practical implications for pitch-side deployment.
- Cluster-based training set filtering as an approach to between-player diversity, with stratified evaluation confirming that prediction difficulty varies systematically by player archetype.

2 Related Research

This section reviews prior work across three areas relevant to this study: physical performance analysis in Gaelic football, machine learning for sports fatigue prediction, and automated time series feature extraction.

Malone et al. [5] analysed high-speed running outputs across the four quarters of elite inter-county Gaelic football matches using GPS data, processed through the standard six-zone velocity threshold framework for Gaelic football. Their study found that HSR distance declined by 19.8% in Q4 relative to Q1, with steeper declines observed in losing teams. Their work validates the quarter partitioning approach and HSR metric definition adopted in this study. Where Malone et al. described Q4 decline retrospectively, this study aims to predict Q4 prospectively using the current game’s Q1–Q3 movement data, each player’s historical Q4 performance across prior matches, and population-level patterns from teammates.

Gamble et al. [4] identified Q4 HSR levels as one of the strongest discriminators between winning and losing inter-county Gaelic football teams. Winning teams consistently sustained their running output in the final quarter while losing teams experienced pronounced declines. This finding emphasises the framing of Q4 prediction as a coaching decision-support problem. The binary framing of teams

either maintaining, or failing to maintain HSR output, motivated the binary Improve or Decline classification target used here. This study aims to predict the rise or fall in HSR in Q4 at an individual player level before Q4 begins, whereas Gamble et al. examined this relationship retrospectively at team level.

Mangan et al. [3] established that players who maintained high levels of HSR in Q4 retained greater passing and technical accuracy, using Pearson’s correlation analysis across 432 player observations from 52 elite matches. This reiterates the importance of HSR as a proxy for overall playing performance in the decisive quarter of a game [4, 11], not merely a fitness indicator, strengthening the case for HSR as the primary prediction target rather than total distance. The between-player variability documented also motivated the use of K-Means clustering to identify player archetypes [5, 15].

Using the same specific dataset, Sheridan et al. [16] demonstrated that GPS-derived action sequences contain learnable patterns predictive of fatigue-related outcomes, a direct methodological precedent for the present study. Their study applied Random Forest and linear regression models to predict Ratings of Perceived Exertion (RPE), a subjective post-match fatigue measure, under a single pooled training configuration [17]. This study aims to predict a physical output measure and explicitly investigates whether models trained on individual player history outperform population-level approaches, a personalised method Sheridan et al. did not address. Fang et al. [18] demonstrated performance drops of 15–20% when machine learning models trained on GPS data from one sport were applied to another, supporting the single-sport design adopted here, arguing against cross-sport generalisation without explicit domain adaptation.

Van Eetvelde et al. [19] analysed machine learning methods in sports performance contexts, finding that gradient boosting and Random Forest approaches consistently outperformed linear models, motivating the inclusion of XGBoost and Random Forest across all experimental conditions. We extend their use of linear models by applying Ridge regression, which adds L2 regularisation appropriate for the small sample size of 99 observations [20].

Jia et al. [14] reviewed 51 deep learning studies in sports performance analysis, finding that LSTM networks were the most frequently applied architecture for temporal sequence data due to their ability to capture long-range dependencies through gated memory mechanisms. Their review also noted that deep learning models require large volumes of labelled data for reliable regression performance. This informed the inclusion of a BiLSTM with attention architecture as a learned sequence baseline, while also setting realistic expectations for its performance relative to simpler tabular models at this data scale. Prior to learned sequence models, classical time series forecasting methods such as ARIMA have been widely applied to sequential sensor data as interpretable baselines [21], motivating their inclusion here as a reference point before introducing neural representations.

Christ et al. [13] introduced TSFresh, a Python package that automatically extracts 794 statistical features from time series data, including autocorrelation, entropy and peak count, combined with False Discovery Rate (FDR) based hypothesis testing for feature selection. Feature selection can be time consuming and domain-specific [13], so their package was introduced to handle the problem of manual feature engineering. We adopted TSFresh as an alternative to our five hand-crafted rolling window features to test whether automated extraction from the 50-step rolling window could outperform interpretable manual features. TSFresh has been applied to a range of datasets [13], but rarely to GPS movement data in field sports. Our study is among the first applications of automated feature extraction from time series data within the GAA, with the closest work coming from [8, 16]. FDR filtering and top-k feature selection were applied inside each LOPO fold using training data only, ensuring the test observation’s label had no influence on which features were selected [13].

Data for this study was collected from 2019–2021, prior to the Football Review Committee introducing their new seven core rule enhancements. The GIU report [22], produced by DCU’s Insight Centre in partnership with the GAA, acknowledged that despite the introduction of these new rules in 2025, the demands for high intensity running have not increased in the way some anticipated. The report also documented significant between-game variability in physical demands [5, 15], corroborating the high between-game and between-player variance observed in our dataset [8]. Where the GIU report is descriptive, this study is predictive, building on these findings.

Collectively, these works establish the motivation for Q4 running performance prediction, validate the GPS data modality and model choices, and identify a clear gap: no prior work has combined individual player history with population-level movement patterns to predict Q4 HSR output before the final quarter begins. This study addresses that gap directly.

3 Methodology

3.1 Data Acquisition and Preprocessing

The dataset was sourced from Antonini et al. [8], comprising GPS tracking data from an anonymised men’s inter-county team of 35 players during 11 games across 3 seasons. Players wore GPS sensors during matches, recording longitude, latitude and speed at 0.1 second intervals. Following the six-zone velocity threshold classification system established as the standard framework for Gaelic football [5], raw coordinates were transformed into a semantically meaningful database of discrete, velocity-banded actions. Each record consists of 12 variables such as action label, duration, game and player.

As per Malone et al. [5] who identified HSR as the metric most sensitive to fatigue-induced decline in Q4 of inter-county Gaelic football matches, HSR metres per minute was selected as the primary metric to characterise running performance [5, 3].

Exploratory analysis confirmed meaningful inter-player variability in Q4 HSR levels (standard deviation = $8.52 \text{ m} \cdot \text{min}^{-1}$, range 2.90–45.79) and a consistent global decline in running performance across the final quarter [5, 15]. These findings motivated the use of personalised training configurations and player archetype clustering [3, 23].

For each player-game combination, 50 rolling 5-minute windows were extracted [8], each stepping back 1 minute from the beginning of Q4 (17.5 minutes into the second half). Combined HSR distance per minute within each window was computed as a univariate sequence input. All 6 action-zone distances per minute were also extracted as a 6-channel multivariate sequence [5]. Five hand-crafted summary statistics were derived from the univariate sequence: mean, standard deviation, final value, overall slope and last-10-step slope. A player-level historical feature was computed for each player-game combination as the mean Q4 HSR per minute across all other games for that player. Historical features were derived from all available games excluding the test game, without enforcing chronological ordering, to maximise usable observations given the limited number of games per player.

Coverage filtering was applied to remove incomplete player-game observations, caused by factors such as substitutions and human operation error [12, 24]. Only instances of players being tracked for a minimum 98% game duration were retained. Outlier removal was performed based on the modified Z-score method of Iglewicz and Hoaglin [25] applied to the ratio of Q4 to Q1–Q3 mean HSR for each player-game combination. After excluding ineligible and anomalous records, 99 player-game observations across 20 unique players remained.

Two prediction tasks were defined. A regression task to estimate Q4 HSR per minute directly, and a binary classification task forecasting whether Q4 output will Improve or Decline relative to Q1–Q3. A ratio threshold of 1.0 created a class distribution of 65 Decline observations (65.7%) and 34 Improve observations (34.3%). This framing captures the nuanced in-game dilemmas facing a coach [4]. Although continuous predictions are more granular, classification holds practical utility to inform real-time tactical decisions [6] by producing a more tractable and actionable signal.

Players were clustered into two archetypes using K-Means ($K = 2$) on three profile features: Q1–Q3 HSR mean, Q4 HSR mean and Q4 HSR standard deviation. The aim was to identify latent similarities in player movement profiles as a proxy for positional and physical attributes not contained in the data [7]. $K = 2$ was selected to promote interpretability and statistically viable cluster sizes, producing clearly defined high-intensity and low-intensity groupings.

3.2 Experiment Design

All experiments followed a strict and identical evaluation protocol. Leave-One-Player-Out (LOPO) cross validation was implemented for all models [16]. This

involves each player-game observation serving as the test set in turn, ensuring no information is leaked from the test game into the training process.

A critical methodological consideration was the computation of the historical feature. For each training row containing the target player, the mean was re-computed to exclude the test game. Players with only two games were therefore excluded from historical feature experiments — holding out the test game and excluding the training game from its own computation leaves no remaining games from which a leakage-free historical value can be derived — leaving 91 observations across 13 players.

Five training configurations were evaluated across all rounds, unless stated otherwise. In all instances, data from the target game was excluded from within-fold training. *All others* excluded the target player and trained on everyone else. *Same player* trained only on the target player’s data. *Combined* used all players, including the target and all others. *Weighted* used the same data as Combined but with the target player upweighted. *Cluster only* limits training input to the target player’s cluster, including their own data. Models trained under all constraints were tested with and without access to the historical input feature.

This design was guided by our research questions. The All others configuration represents the cold-start scenario: a player who has not been previously seen by the model must rely entirely on population-level movement patterns. The other four settings explore how previous knowledge of a player’s performance may improve predictions. Comparisons between inclusion and exclusion of the historical feature allow analysis of how performance is enhanced by access to established fatigue profiles versus generalisation to completely new environments.

All models were initially evaluated using default hyperparameters to enable consistent comparisons across experimental rounds. Hyperparameter tuning was subsequently applied to selected headline models.

Model performance was evaluated using Mean Absolute Error (MAE) and R^2 for regression, and weighted F1-score and accuracy for binary classification [16]. Weighted F1 was selected as the primary classification metric to account for the class imbalance between Decline (65.7%) and Improve (34.3%) observations.

Round 0: Naive Baseline. Two rule-based naive baselines were established as a benchmark for all subsequent comparisons. The regression baseline predicts the current game Q1–Q3 HSR mean as the Q4 value. The classification baseline always predicts the majority class, Decline. Any model that cannot significantly outperform these baselines offers little to no predictive power or practical value.

Round 1: ARIMA and ARIMAX. The first model-based experiment framed Q4 prediction as a classical time series forecasting problem [21]. Each player-game’s 50-step HSR sequence was treated as a univariate time series, with `auto_arima` selecting optimal order and differencing based on AIC. Visualisations suggested a potential seasonal component within quarters, however SARIMA

was ruled out after ACF analysis. Four approaches were evaluated using 18 forecast steps: equal-weighted averaging, exponentially-weighted averaging, and ARIMAX variants of both incorporating the historical feature as an exogenous variable. This trial establishes whether simple temporal extrapolation of pre-Q4 patterns can predict Q4 performance.

Round 2: BiLSTM with Attention. The second round replaced classical forecasting with a learned sequence model. A bidirectional LSTM with single-head additive attention was trained on the full 6-channel multivariate sequence across 50 timesteps [14]. Separate specialised models were trained for regression and classification per fold. For historical experiments, the additional feature was concatenated to the attended representation before the output layer. The LSTM was evaluated across all configurations except Weighted, as reliable sample up-weighting could not be implemented without significant engineering beyond the scope of this project.

Round 3: Hand-Crafted Tabular Models. The third round shifted focus to interpretable summary statistics. Five hand-crafted features were derived from the univariate HSR sequence [19]. Ridge regression, RF, and XGBoost were trained on these features with and without the historical feature. This round tests whether domain-driven interpretable features can match or exceed the performance of the sequence models from Round 2.

Round 4: TSFresh Automated Feature Engineering. The final round mirrored Round 3 but replaced hand-crafted inputs with automatically extracted features using the TSFresh library. Features were extracted from each sequence individually. Round 3’s feature budget was matched by selecting 5 features within each fold using FDR filtering followed by Ridge coefficient importance ranking [13]. This round investigated whether automated features could establish a new performance ceiling over hand-crafted engineering.

4 Results and Discussion

4.1 Results

The regression target variable, Q4 HSR per minute, has a mean of $26.35 \text{ m}\cdot\text{min}^{-1}$ across the 99 eligible player-game observations. The naive baseline produces MAE $5.779 \text{ m}\cdot\text{min}^{-1}$, representing a prediction error of 21.9% of the mean. The benchmark classification achieved by always predicting the majority class (Decline, 65.7%) is F1 0.520. These metrics represent the minimum threshold any learned model must surpass.

Table 1 presents selected LOPO evaluation results across all four rounds of experimentation, with the best cold-start and personalised results displayed per model family.

Table 1. Selected LOPO results across all rounds. Best cold-start and personalised results shown per model family.

Round	Model	Config	MAE	R^2	F1
0	Naive Baseline	—	5.779	—	0.520
1	ARIMA	—	6.729	-0.092	0.576
1	ARIMAX	—	5.893	0.185	0.537
2	LSTM (mv+hist)	All others	6.223	—	0.714
2	LSTM (mv)	Combined	—	—	0.742
3	Ridge (hc+hist)	All others	5.090	—	—
3	Ridge (hc+hist)	Cluster only	4.820	—	0.703
3	RF (hc+hist)	Weighted	4.914	0.453	—
4	Ridge (ts+hist)	All others	5.559	—	—
4	Ridge (ts+hist)	Cluster only	4.732	0.507	—
4	Ridge (ts)	Cluster only	—	—	0.718

Round 1: ARIMA and ARIMAX. Pure ARIMA (MAE 6.729, $R^2 = -0.092$) performs significantly below the naive regression baseline. This confirms that simple, temporal extrapolation of the pre-Q4 sequence cannot predict Q4 outcomes. A negative R^2 further compounds that simply predicting the mean leads to better predictions. Clearly, the model infers a trend that does not continue into Q4. Exponential weighting of the 18-step forecast, which prioritises earlier steps based on the assumption that they would be more accurate, fails to yield an improvement (MAE 6.893 vs 6.729). This suggests the forecasted values are uniformly unreliable rather than progressively declining in relevance. ARIMAX, incorporating the player’s historical Q4 average as an exogenous variable, improves substantially on regression (MAE 5.893, $R^2 = 0.185$) but performs worse on classification than pure ARIMA (F1 0.537 vs 0.576). Although the historical average provides a useful magnitude anchor for regression, it shifts predictions towards that long-term mean and suppresses the within-game directional signal that determines Q4 improvement or decline.

Round 2: BiLSTM with Attention. The LSTM significantly outperforms ARIMA across both tasks, validating the use of learned sequence representations. Under cold-start conditions, the LSTM with historical features achieves F1 0.714, a 0.138 gain over ARIMAX. With personalised Combined training, the LSTM without historical features gives F1 0.742, the strongest classification result in the pre-tuning experiments. However, regression performance is still particularly poor — Others only MAE 6.223 fails to improve on the naive baseline. These findings reveal a task-specific divergence. The multivariate temporal structure captured by the LSTM provides a strong classification signal but does not translate to precise HSR distance estimation.

Round 3: Hand-Crafted Features. Ridge regression with the five hand-crafted features and the historical feature achieves MAE 5.090 under cold-start conditions. This is a considerable gain (1.133) on the LSTM equivalent at a fraction of the computational cost. Under personalised Cluster only training with the same features, Ridge gives strong scores of MAE 4.82 and F1 0.703. RF produces the best Weighted regression result across the entire study (MAE 4.914, $R^2 = 0.453$), demonstrating that ensemble methods can offer meaningful gains in the personalised setting. XGBoost, however, consistently underperforms the other classical ML models across all configurations despite identical feature sets and training splits. This suggests that gradient boosting’s complexity is not well matched to this dataset’s limited scale.

Round 4: TSFresh Automated Features. Similarly to Round 3, Ridge led the way for regression performance among the three classical models. When trained using Cluster only data, the addition of the historical mean to the TSFresh features resulted in the lowest MAE across all pre-tuning experiments (MAE 4.732, $R^2 = 0.507$). However, the TSFresh cold-start regression (MAE 5.559) is markedly less impressive than its hand-crafted counterpart (MAE 5.090). Automated features clearly require the context of personalised training to outperform domain-guided features. Classification performance of TSFresh models is consistently weaker than the best in class LSTM across all configurations (best TSFresh F1 0.718 vs LSTM 0.742), thus reinforcing the conclusion that raw sequence structure carries inherently unique, classification-relevant signalling.

Hyperparameter Tuning. Following the main experiments, four headline models were selected and tuned. These were the best performing regression and classification algorithms under cold-start and personalised conditions. Ridge alpha regularisation was tuned across a grid of 0.001–100, identifying $\alpha = 10$ as optimal for both regression models. For classification, LSTM architecture was tuned across hidden size, dropout, and learning rate, identifying hidden= 32, dropout= 0.3, lr= 0.001 for the cold-start model and hidden= 16, dropout= 0.5, lr= 0.002 for the personalised model. Tuned headline model performance with 95% bootstrap confidence intervals is presented in Table 2. The overlapping confidence intervals between scenarios indicate the improvement with personalisation is consistent but not statistically definitive at this sample size.

Table 2. Tuned headline model performance with 95% bootstrap confidence intervals.

Model	Config	MAE (95% CI)	F1 (95% CI)
Ridge (hc+hist)	All others	4.883 [4.255, 5.794]	—
Ridge (ts+hist)	Cluster only	4.721 [4.106, 5.520]	—
LSTM (mv+hist)	All others	—	0.743 [0.658, 0.826]
LSTM (mv)	Combined	—	0.763 [0.678, 0.848]

4.2 Discussion

Supplementary visualisations supporting the following analysis are provided in Appendix A.

Our first research question investigated whether pre-Q4 movement patterns can predict Q4 high-speed running performance. All learned models improve on the naive baselines for both regression and classification, confirming that pre-Q4 movement patterns carry genuine predictive signal. The best regression model (Ridge, TSFresh+hist, Cluster only, MAE 4.721 ± 0.721) reduces prediction error to 17.9% of the target variable mean, a 4% improvement compared to the baseline 21.9%. The best classification model (LSTM, Combined, F1 0.763 ± 0.085) achieves a meaningful gain over the complete majority voting (F1 0.520).

ARIMA’s failure to improve on baseline performance is theoretically significant. It confirms that Q4 represents a structural break in player movement behaviour rather than merely a continuation of early game trends [5]. This makes classical extrapolation techniques fundamentally unsuitable. Learned representations of the full pre-Q4 window are required.

Another key finding is the task-specific divergence between regression and classification performance across model families. Ridge hand-crafted models excel on regression while LSTM wins on classification. Predicting the magnitude of Q4 HSR values is well-served by compact statistical summarisation of the pre-Q4 sequence, but predicting the direction of change requires the temporal and cross-zone structure captured from LSTM’s full multivariate sequence. In practical terms, regression and classification are not interchangeable tasks and benefit from different model architectures.

XGBoost consistently underperforms simpler models across all configurations and contexts. This is a common finding in small-dataset settings. The bias-variance tradeoff favours regularised linear models and moderate ensembles over gradient boosting’s high capacity [19], reinforcing that dataset scale is a binding constraint on model complexity.

Ridge regression is the most well-suited to this setting of the three classical machine learning models. Its L2 regularisation explicitly penalises large coefficients, preventing overfitting to the small training folds produced by LOPO. Despite its ensemble averaging, Random Forest appears less robust to high variance given the limited training sets.

Table 3 shows that the availability of the target player’s data during training consistently improves predictions, directly answering our second research question. Regression performance increases across every model family with the exception of ARIMA. The largest gains are seen in RF (TSFresh+hist, Weighted, $\Delta\text{MAE} -1.524$) and LSTM (mv+hist, Combined, $\Delta\text{MAE} -0.999$) where both models

start from weak cold-start baselines where population-level training alone is insufficient. Ridge (hc+hist, Cluster only, $\Delta\text{MAE} -0.270$) shows the smallest gain as it already achieves the strongest cold-start regression, leaving less room for improvement. Classification enhancement is more modest but consistent nonetheless (average $\Delta\text{F1} +0.036$ excluding ARIMA). Undoubtedly, personalisation benefits both tasks although regression certainly improves more substantially.

Table 3. Cold-start vs personalised training – best result per model family. Cold start = all_others split. Personalised = best of same_player, combined, weighted, cluster_only. MAE in $\text{m}\cdot\text{min}^{-1}$.

Model	Features	Cold MAE	Cold F1	Best personal. split	Pers. MAE	Pers. F1
ARIMA	—	6.729	0.576	forecast+hist	5.893	0.537
LSTM	mv+hist	6.223	0.714	combined	5.224	0.742
Ridge	hc+hist	5.090	0.689	cluster_only	4.820	0.707
RF	hc+hist	5.928	0.667	weighted	4.914	0.708
XGBoost	hc+hist	6.387	0.672	cluster_only	5.559	0.721
Ridge	tsfresh+hist	5.559	0.648	cluster_only	4.732	0.740
RF	tsfresh+hist	6.503	0.609	weighted	4.979	0.704

Contrary to expectation, across all four headline models, no statistically significant relationship was found between the number of historical games available and prediction accuracy. Spearman correlations ranged from $\rho = -0.448$ to $\rho = +0.086$ for regression MAE and $\rho = -0.098$ to $\rho = +0.286$ for classification accuracy, with no model reaching significance (all $p > 0.10$). The 7–8 game range consistently produces the highest error across both regression models. This suggests that within the 3–11 game range available in this dataset, the consistency of a player’s Q4 profile matters more than the volume of prior observations.

This finding contextualises the underperformance of the Same player split relative to Combined, Weighted and Cluster only. If more games reliably improved prediction, the same player would be expected to approach combined performance as player history grows. The absence of this relationship suggests that, at the current scale, a wider population context remains necessary. The implication for practitioners is that collecting data from a wider cohort of players may be more valuable than accumulating additional games for individual players, at least until per-player game counts reliably exceed the range observed here.

A series of feature importance and ablation studies were conducted to understand the headline models’ decision making, in line with our final research question. Ablation analysis of Ridge (hc+hist, All others) reveals a clear hierarchy of feature importance. Removing the historical features increases MAE by $0.634 \text{ m}\cdot\text{min}^{-1}$. This is by far the largest single feature impact and exceeds the combined effect

of removing each hand-crafted sequence feature. Knowing a player’s typical Q4 running performance is more informative than any pattern derived from their current game’s movement alone.

series_mean contributes most meaningfully ($\Delta\text{MAE} +0.065$ when removed) of all hand-crafted features and interestingly has the highest Ridge coefficient of all features, including the historical mean. This intuitively reveals that a player’s average pre-Q4 intensity is the best single-game predictor of their Q4 output.

Unexpectedly, removing the series standard deviation and slope as features actually improved performance ($\Delta\text{MAE} -0.054$ and -0.073 respectively), indicating that these features introduce noise rather than signal at this scale. These results motivated a refined model with those two features removed. This improved on the original performance across both training configurations and both tasks (cold-start: MAE 4.883 vs 5.003, F1 0.697 vs 0.689; combined: MAE 4.743 vs 4.876, F1 0.742 vs 0.723).

Table 4. Refined Ridge HC (3 features + hist) vs original (5 features + hist) with 95% bootstrap CIs. Ablation-motivated refinement removing series_std and series_slope. MAE in $\text{m}\cdot\text{min}^{-1}$.

Model	Split	n	MAE [95% CI]	R^2 [95% CI]	F1 [95% CI]
Original (5 feats)	all_others	91	5.000 [4.255, 5.794]	0.426 [0.205, 0.585]	0.689 [0.585, 0.782]
Refined (3 feats)	all_others	91	4.873 [4.173, 5.638]	0.462 [0.267, 0.607]	0.697 [0.600, 0.790]

For Ridge (TSFresh+hist, Cluster only), fold-frequency analysis of feature selection across 91 LOPO folds revealed one particularly dominant feature: value__c3__lag_3. Selected in 65.9% of folds, the non-linear autocorrelation statistic consistently ranked first in importance. This aligns with the theoretical expectation that within a homogeneous intensity-level cluster, the directional momentum of a player’s pre-Q4 HSR sequence may be more discriminative than its absolute magnitude. A feature count sweep from $k = 1$ to $k = 10$ showed that $k = 1$ alone achieves MAE 4.899. Performance never meaningfully progresses past the original $k = 5$, showing that the predictive signal is concentrated in a small number of the automated TSFresh features rather than distributed across a large set.

The Cluster only training configuration producing the best regression result in the entire study directly validates the clustering methodology [5, 15]. The model benefits from retaining its personalised context within a wider network of similar players. Cluster only models consistently outperforming those trained on the full population (Combined, Weighted) and solely the target player (Same player) provides strong evidence that the K-Means clustering successfully iden-

tified latent similarities between players that are genuinely predictive of their Q4 running performance.

LSTM permutation importance reveals a meaningful contrast between training configurations. Under personalised Combined training, High Intensity Running (+0.015 F1) is the most informative zone. However, under cold-start, the same activity band is actively harmful (-0.013 F1) [5, 15]. This reveals an interesting concept. High-intensity zone dynamics are player-specific. The same HIR pattern in Q1–Q3 means different things for different players in terms of Q4 impact. Personalised training allows the LSTM to learn these individual patterns in a way that population-level training cannot generalise as reliably.

A central finding to this study is that the performance gap between the most interpretable model and the best-performing model is small in practical terms. The refined Ridge (hc+hist, All others) model with 4 fully interpretable features achieves MAE 4.883 [4, 26]. This constitutes a gap of $0.162 \text{ m}\cdot\text{min}^{-1}$ relative to the best TSFresh model. Given the target variable mean of $26.35 \text{ m}\cdot\text{min}^{-1}$, these differences are negligible for substitution or tactical decisions.

TSFresh feature selection is fold-dependent, requiring the full training dataset at inference time, and its features are domain agnostic and complex. Conversely, Ridge (hc, All others) can be explained to a practitioner in a single sentence: “We consider the player’s average pre-Q4 intensity, their intensity in the final window, their recent trend, and their historical Q4 average.”

Similarly, for real-time binary classification where a coach requires a compelling justification, the LSTM’s (mv, Combined) opaque attention mechanism provides no such reasoning. The F1 advantage over Ridge (hc, All others) (0.763 vs 0.742) may not justify the loss of interpretability in a deployed setting.

Furthermore, the refined Ridge (hc+hist, All others) model holds a distinct practical advantage in cold-start scenarios. It requires no prior knowledge of the target player, making it immediately applicable to newly observed players or teams where previous data is unavailable. Although Ridge (TSFresh+hist, Cluster only) and LSTM (mv, Combined) are the technically superior choice, for pitch-side decision support, Ridge (hc+hist, All others) is the recommended practical configuration for both regression and classification.

This study has several limitations. The dataset comprises 99 observations from a single inter-county team across three seasons, which restricts the generalisability of findings across teams, competitions, and playing styles. GPS devices have known accuracy limitations [24], with measurement error particularly pronounced during higher-intensity activities [27]. The introduction of new GAA playing rules since data collection may also affect the applicability of learned movement patterns to the current real-world context [22].

Future work should prioritise the construction of a real-time data pipeline following the approach of Antonini et al. [8], enabling in-game prediction during matches rather than retrospective analysis. Expanding the dataset across multiple teams and seasons would allow investigation of how model performance scales with data volume. Potential areas for improvement lie within cold-start deployment as the population prior becomes more representative and under Same player training with significantly richer player histories. A larger cohort would also support more granular clustering, potentially recovering positional archetypes rather than broad intensity groupings. This may further improve the Cluster only personalisation strategy that already proved effective here.

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Appendix A

Figure 1: Q4 HSR Distribution and Inter-Player Variability

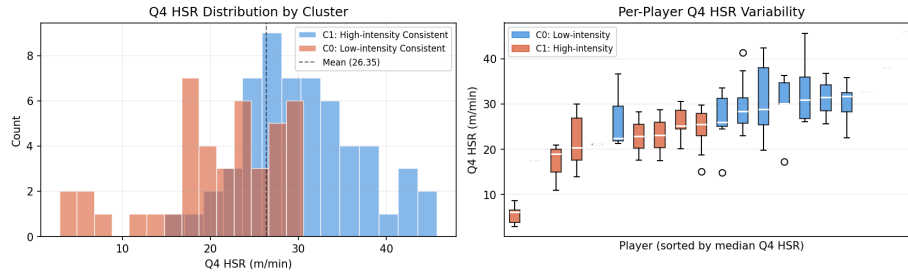


Figure 2: Mean Pre-Q4 HSR Sequence by Cluster

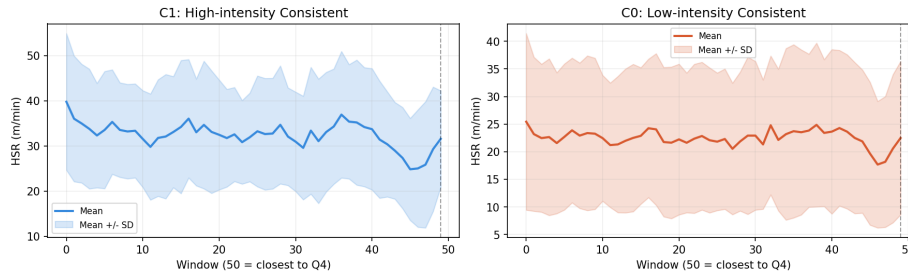


Fig. 2. Mean Pre-Q4 HSR Sequence by Cluster

Figure 3: Feature Ablation -- Ridge hc+hist all_others

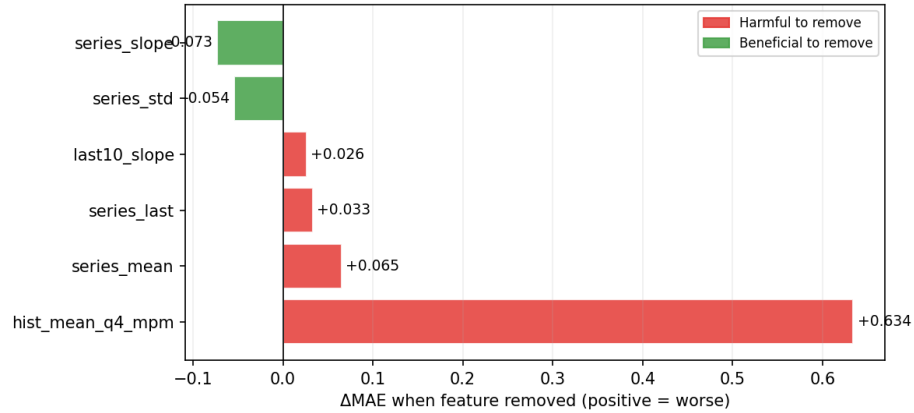


Fig. 3. Feature Ablation – Ridge hc+hist all_others

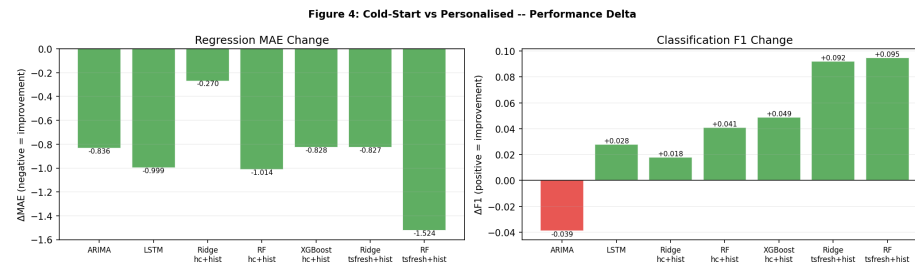


Fig. 4. Cold-Start vs Personalised – Performance Delta

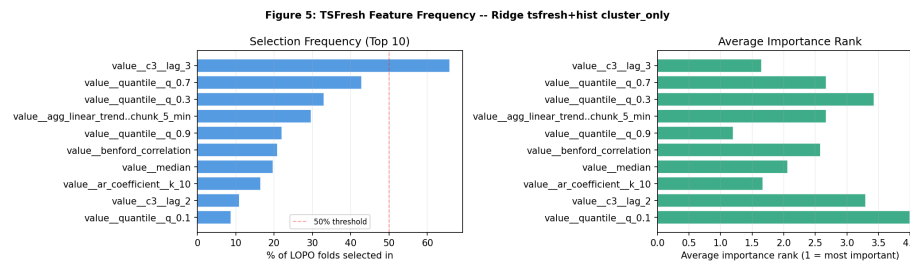


Fig. 5. TSFresh Feature Frequency – Ridge tsfresh+hist cluster_only

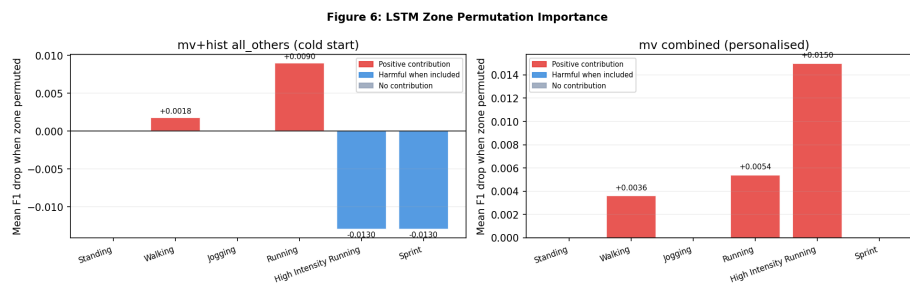


Fig. 6. LSTM Zone Permutation Importance

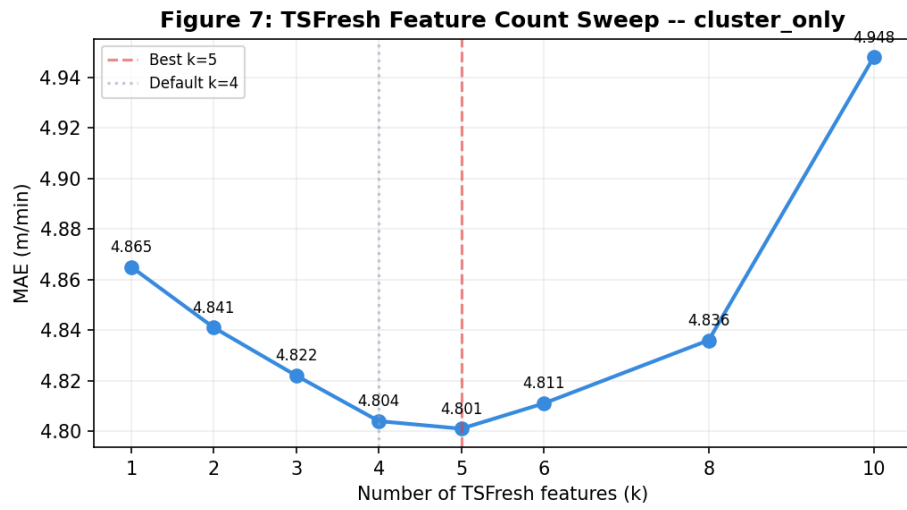


Fig. 7. TSFresh Feature Count Sweep – cluster_only